O–W–41a
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SUPERSEDING
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17 March 1952

FEDERAL SPECIFICATION

WATER, BATTERY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers battery water for use in storage batteries.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications.—There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

(Non-Government activities may obtain copies of Federal Specifications and Standards upon application accompanied by check, money order, cash, or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets SW., Washington 25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Copies of the Index of Federal Specifications and Standards may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C. Prices may be obtained from the Index of Federal Specifications and Standards or from the GSA Regional Offices.

(Single copies of this specification and other product specifications required by non-Government activities for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards through departmental channels from the established sources of supply.)

2.2 Standard.—The following Military Standard, of the issue in effect on date of invitation for bids, forms a part of this specification:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.3 Other publications.—The following publications, of the issues in effect on date of invitation for bids, unless otherwise stated, form a part of this specification:

Consolidated Classification Committee:
Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

American Trucking Association, Inc.:
Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, Inc., 1424 16th Street, NW., Washington 6, D. C.)

3. REQUIREMENTS

3.1 Physical characteristics.—Battery water shall be clear and free from suspended matter, color, odor, and taste (see 4.4.1).

3.2 Impurities.—When tested in accordance with the applicable method specified in 4.4, the battery water shall conform to the requirements shown in table I.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—For purposes of sampling, a lot shall consist of all battery water manufactured as one batch, and offered for delivery at one time.

4.1.2 *Sampling for inspection of filled containers.*—A random sample of filled containers shall be taken from each lot by the Government inspector in accordance with Military Standard MIL–STD–105 at Inspection Level I and Acceptable Quality Level (A. Q. L.) equal 2.5 per-

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TABLE I.—*Impurities*

Requirement	Percent maximum	Test paragraph
Total solids	0.01	4.4.2
Organic and volatile matter	.005	4.4.3
Calcium and magnesium as CaO	.004	4.4.4
Iron	.00005	4.4.5
Copper	.00025	4.4.6
Chloride	.0005	4.4.7
Nickel	.00002	4.4.8
Ammonia as NH_4	.0008	4.4.9
Nitrites as NO_2	.0005	4.4.10
Nitrates as NO_3	.001	4.4.11

cent defective to verify compliance with this specification regarding fill, closure, marking and other requirements not involving tests.

4.1.3 *Sampling for tests.*

4.1.3.1 When offered for delivery in tank cars or wagons, three one-gallon samples shall be taken at point of delivery, from each tank, one from near the top, one from near the middle, and one from near the bottom of the battery water.

4.1.3.2 When offered for delivery in carboys or other glass containers, one-gallon samples shall be taken from three containers out of each sub-lot of not more than 5,000 gallons.

4.1.3.3 The three individual samples from each tank (see 4.1.3.1) or from each sub-lot containers (see 4.1.3.2) shall be sealed and labeled by the Government inspector and forwarded to the Government laboratory designated by the procuring activity for test purposes.

4.2 Inspection.

4.2.1 *Inspection of filled containers.*—Each sample filled container selected in accordance with 4.1.2 shall be examined by the Government inspector for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects, or under the required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard MIL-STD-105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or repaired all nonconforming containers.

4.3 Lot acceptance tests.—The sample specimens selected in accordance with 4.1.3 shall be subjected to the tests specified in 4.4, separately. If either specimen fails one or more of these tests the lot shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or reworked all nonconforming products.

4.4 Tests.

4.4.1 *Physical charteristics.*—A well shaken sample shall be poured into a clean, dry glass vessel, and examined for sediment, turbidity, odor, color, and taste.

4.4.2 *Total solids.*—Evaporate to dryness 500 ml. of the well shaken sample in a tared platinum evaporating dish on a steam bath and subsequently dry to constant weight at 105° ±2° C. Cool in a desiccator and reweigh. Calculate percentage total solids as follows:

$$\text{Percentage total solids} = \frac{\text{Weight of residue (grams)}}{5}$$

4.4.3 *Organic and volatile matter.*—The residue obtained in the determination for total solids is ignited at a low red heat. Cool in a desiccator and reweigh. Calculate the organic and volatile matter as follows:

$$\text{Percentage organic and volatile matter} = \frac{A-B}{5}$$

where:

A = Weight of residue obtained as specified in 4.4.2

B = Weight of residue obtained as specified in 4.4.3

4.4.4 *Calcium and magnesium oxide.*

4.4.4.1 *Reagents.*

4.4.4.1.1 *Standard calcium chloride solution.*—Weigh 0.446±0.001 gm. of calcium carbonate into a 400-ml. beaker. With the beaker covered with a watch glass, cautiously add a small quantity of diluted HCl until the calcium carbonate is dissolved. Dilute with distilled water to about 350 ml. and neutralize with diluted NH_4OH to a slight alkalinity, using litmus as the indicator. Transfer to a 500-ml. volumetric flask, carefully washing the beaker

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with distilled water and dilute to the mark. Stopper and agitate. This solution has an equivalent concentration of 0.0005 gm. of CaO per milliliter.

4.4.4.1.2 *Standard soap solution.*—Dissolve 10 gm. of pure castile soap in a liter of 95 percent ethyl alcohol. Dilute a portion of this solution with 95 percent ethyl alcohol until approximately 1 ml. of diluted soap solution is required to titrate a solution of 2 ml. of the standard calcium carbonate solution diluted to 50 ml. with distilled water. (See 4.4.4.2 for the titrating procedure.) Note the volume of diluted soap solution needed to titrate the diluted calcium carbonate solution.

4.4.4.2 *Procedure.*—Adjust the alkalinity of the sample so that it is just pink to phenolphthalein. Place 50 ml. of the sample in an 8-ounce square bottle provided with a stopper. Titrate with standardized soap solution; adding solution in small amounts and shaking vigorously after each addition until a permanent lather is obtained. A permanent lather is defined as one that completely covers the surface of the sample, with the bottle on its side, for at least 5 minutes. Note number of milliliters of standardized soap solution required.

4.4.4.3 *Lather factor (blank).*—Place 50 ml. of CO_2-free special distilled water in an 8-ounce bottle provided with a stopper. Titrate with standardized soap solution repeating procedure specified in 4.4.4.2.

4.4.4.4 *Calculations.*—Calculate the calcium-magnesium content as calcium oxide as follows: Calcium and magnesium as *CaO*, percentage=

$$\left(\frac{B-L}{A-L}\right)\left(\frac{1}{500}\right)$$

where:

A=ml. diluted soap solution noted in 4.4.4.1.2
B=ml. diluted soap solution noted in 4.4.4.2
L=lather factor noted in 4.4.4.3

4.4.5 *Iron.*

4.4.5.1 *Reagents.*

4.4.5.1.1 *Special distilled water.*—Special distilled water, free from iron, copper, and nickel, shall be used in preparing the required reagent solutions and in making the dilutions indicated in the test procedures. Water which has been double-distilled in all-glass equipment is considered suitable for this purpose.

4.4.5.1.2 *Stock iron solution.*—Dissolve 0.70 gm. of crystallized ferrous ammonium sulphate, $FeSO_4(NH_4)_2\ SO_4{\cdot}6H_2O$ in 50 ml. of special distilled water and carefully add 5 ml. reagent grade sulfuric acid (sp. gr. 1.84). To the warm solution add potassium permanganate solution (see 4.4.5.1.5) until a pink color persists for 1 minute. Dilute to 1 liter with special distilled water.

4.4.5.1.3 *Standard iron solution.*—Dilute 5 ml. of the stock solution to 100 ml. with special distilled water. (This solution should be prepared for immediate use only.) This solution contains 0.005 mg. Fe per ml.

4.4.5.1.4 *Potassium thiocyanate solution.*—Dissolve 120 gm. of potassium thiocyanate in special distilled water and dilute to 1 liter.

4.4.5.1.5 *Potassium permanganate solution.*—Dissolve 6.3 gm. of potassium permanganate in special distilled water and dilute to 1 liter.

4.4.5.2 *Procedure.*—Fill one of two matched 50-ml. Nessler tubes to the mark with the sample. Into the other Nessler tube, pipette 5 ml. of standard iron solution and fill to mark with special distilled water. To each tube add 5 ml. reagent grade hydrochloric acid (sp. gr. 1.18). To each tube add one drop of potassium permanganate solution at a time until a pink color persists for 1 minute. Then add 5 ml. of potassium thiocyanate solution to each tube, shake well, and compare the colors produced by looking down into the tubes. If the color produced by the sample is darker than that of the standard the distilled water contains more than 0.0005 percent iron.

4.4.6 *Copper.*

4.4.6.1 *Reagents.*

4.4.6.1.1 *Special distilled water.*—See 4.4.5.1.1.

4.4.6.1.2 *Standard copper solution.*—Dilute 25 ml. of a stock solution of copper sulfate (1.966 gm. chemically pure $CuSO_4{\cdot}5H_2O$ in 1 liter of special distilled water) to 500 ml. with special distilled water. This solution contains 0.025 mg. Cu per ml.

4.4.6.1.3 *Sodium diethyldithiocarbamate solution.*—Dissolve 0.5 gm. of sodium diethyldithiocarbamate in 500 ml. of special distilled water and keep in a dark glass bottle protected from light.

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4.4.6.2 *Procedure.*—Fill one of two matched 50-ml. Nessler tubes to the mark with the sample. Into the other Nessler tube, pipette 1 ml. of standard copper solution and fill to mark with special distilled water. Into each Nessler tube add 1 ml. of ammonium hydroxide (sp. gr. 0.90). Thoroughly mix and then add 2 ml. of sodium diethyldithiocarbamate solution. Stopper both tubes and shake well. Compare the colors produced by looking down into the tubes. If the color produced by the sample is darker than that of the standard, the distilled water contains more than 0.00025 percent copper.

4.4.7 *Chloride.*

4.4.7.1 *Reagents.*

4.4.7.1.1 *Silver nitrate solution.*—Dissolve 2.0 gm. of silver nitrate in 100 ml. of special distilled water.

4.4.7.1.2 *Standards chloride solution.*—Dissolve 0.1648 gm. of fused sodium chloride in special distilled water and make up to 1 liter. This solution contains 0.1 mg. Cl in 1 ml.

4.4.7.2 *Procedure.*—Fill one of two matched 50-ml. Nessler tubes to the mark with the sample. Into the other Nessler tube, pipette 2.5 ml. of standard chloride solution and dilute to the mark with special distilled water. Into each tube add 1 ml. of 1:10 nitric acid and 1 ml. of the silver nitrate solution. Compare the turbidity produced by looking down into the tubes. If the turbidity produced by the sample is greater than that of the standard, the water contains more than 0.0005 percent chloride.

4.4.8 *Nickel.*

4.4.8.1 *Reagents required.*

4.4.8.1.1 *Special distilled water.*—See 4.4.5.1.1.

4.4.8.1.2 *Standard nickel solution.*—Dilute 10 ml. of a stock solution of nickel chloride (0.405 gm. chemically pure $NiCl_2 \cdot 6H_2O$ in 1 liter of special distilled water) to 100 ml. with special distilled water.

4.4.8.1.3 *Dimethylglyoxime solution.*—Dissolve 0.5 gm. of dimethylglyoxime in 500 ml. of 95 percent of ethyl alcohol and keep in a dark glass bottle protected from light.

4.4.8.2 *Procedure.*—Fill one of two matched 50 ml. Nessler tubes to the mark with the sample. Into the other Nessler tube, pipette 1 ml. of standard nickel solution and fill to mark with special distilled water. Into each Nessler tube add saturated bromine water, drop by drop, shaking well after the addition of each drop, until a residual color of bromine is observed in the tubes. Then add ammonium hydroxide (sp. gr. 0.90), drop by drop, into each tube until the bromine color is removed. Add 0.5 ml. of dimethylglyoxime solution, stopper both tubes and shake well. Compare the colors produced by looking down into the tubes. If the color produced by the sample is darker than that of the standard, the distilled water contains more than 0.00002 percent nickel.

4.4.9 *Ammonia as NH_4.*

4.4.9.1 *Reagents.*

4.4.9.1.1. *Nessler reagent.*—Dissolve 61.75 gm. of potassium iodide in 250 ml. of special distilled water, and add a cold solution of mercuric chloride which has been saturated by boiling with an excess of mercuric chloride. Pour in the chloride solution cautiously, adding an amount just sufficient to develop a permanent bright red color (approximately 400 ml. of saturated mercuric chloride solution will be required). Dissolve the red precipitate by adding exactly 0.75 gm. of potassium iodide. Add 150 gm. of potassium hydroxide dissolved in 250 ml. of water. Make up to 1 liter. Mix thoroughly and allow the precipitate to settle. Use the clear supernatant liquid.

4.4.9.1.2 *Standard ammonium chloride solution.*—Dilute 10 ml. of a stock solution (2.9656 gm. of ammonium chloride dissolved in 1 liter of special distilled water) to 100 ml. with special distilled water. This solution contains 0.1 mg. NH_4 per 1 ml. of solution.

4.4.9.1.3 *Zinc sulfate solution.*—Dissolve 100 gm. of zinc sulfate ($ZnSO_4 \cdot 7H_2O$) in special distilled water and dilute to 1 liter.

4.4.9.1.4 *Sodium hydroxide.*—Dissolve 50 gm. of sodium hydroxide in special distilled water and make up to 100 ml.

4.4.9.2 *Procedure.*—Add 1 ml. of the zinc sulfate to 100 ml. of sample. Adjust the alkalinity by adding 0.4 to 0.5 ml. of the sodium hydroxide solution to a pH value of 10.5. Mix and allow to stand for several minutes. Filter or centrifuge. Use the clear liquid for test. Fill one of the two matched 50-ml. Nessler tubes to the mark with the sample. Into the other

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Nessler tubes, pipette 4 ml. of the standard ammonium chloride solution and dilute to mark with special distilled water. Into each Nessler tube add 1 ml. of Nessler reagent and allow to stand for at least 10 minutes. Compare the colors produced by looking down into the tubes. If the color produced by the sample is darker than that of the standard, the battery water contains more than 0.0008 percent ammonia as NH_3.

4.4.10 *Nitrites as NO_2.*

4.4.10.1 *Reagents.*

4.4.10.1.1 *Sulfanilic acid solution.*—Dissolve 8 gm. of ACS grade sulfanilic acid in 1 liter of 5 *N* acetic acid (sp. gr. 1.041, or 285 ml. glacial acetic acid diluted to 1 liter).

4.4.10.1.2 *Alpha-naphthylamine-acetate solution.*—Dissolve 5 gm. of solid alpha-naphthylamine in 1 liter of 5 *N* acetic acid. Filter the solution.

4.4.10.1.3 *Sodium nitrite stock solution.*—Dissolve 0.150 gm. of ACS grade sodium nitrite in one liter of distilled water.

4.4.10.1.4 *Standard sodium nitrite solution.*—Dilute 50 ml. of sodium nitrite stock solution to 1000 ml. with distilled water.

4.4.10.2 *Procedure.*—Add 1 ml. of sulfanilic acid solution to 100 ml. of sample and 100 ml. of standard sodium nitrite. Allow the two solutions to stand for 5 minutes and then add 1 ml. of the alpha-naphthylamine solution to each. Agitate thoroughly and allow to stand for 10 minutes. Compare the color of the sample with that of the standard using two matched 50-ml. Nessler tubes. If the color of the sample is darker than that of the standard, the battery water contains more than 0.0005 percent nitrite as NO_2.

4.4.11 *Nitrates as NO_3.*

4.4.11.1 *Reagents.*

4.4.11.1.1 *Phenoldisulfonic acid solution.*—Dissolve 25 gm. of pure white phenol in 150 ml. of pure concentrated sulfuric acid, add 75 ml. of fuming sulfuric acid (15 percent SO_3), stir well and heat for 2 hours at about 100° C.

4.4.11.1.2 *Standard nitrate solution.*—Dissolve 0.8150 gm. of ACS grade potassium nitrate in 1 liter of special distilled water. Evaporate 100 ml. of this solution to dryness on a water bath. Moisten the residue quickly with 5 ml. of phenol disulfonic acid solution and macerate by rubbing with a glass rod to insure intimate contact. Dilute to 500 ml. One ml. of this standard solution contains 0.1 mg. of NO_3.

4.4.11.2 *Procedure.*—Evaporate 100 ml. of the sample to dryness on a steam bath. Cool and add 2 ml. of phenol disulfonic acid solution and macerate residue by rubbing with a glass rod. Add 20 ml. of special distilled water, then neutralize with 20 ml. of ammonium hydroxide (1:1). Dilute to 50 ml. Fill one of two matched 50 ml. Nessler tubes to the mark with the sample. To the other Nessler tube, pipette 5 ml. of standard nitrate solution; add 20 ml. of ammonium hydroxide (1:1) and dilute to the 50-ml. mark with special distilled water. Compare the colors by looking down into the tubes. If the color of the sample is darker than that of the standard, the battery water contains more than 0.001 percent NO_3.

4.5 Rejection.—The Government inspector shall reject a tank or sub-lot if he receives an unfavorable report on any sample from the test laboratory.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified, packaging shall be in accordance with manufacturer's commercial practice.

5.2 Packing.—Unless otherwise specified, the subject material shall be packed in accordance with the Consolidated Freight Classification Rules, or Motor Freight Classification Rules, whichever may be applicable.

5.3 Marking.—Unless otherwise specified, interior and exterior shipping containers shall be marked with the name of the material as defined by the contract or order under which shipment is made, the quantity contained therein, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 Ordering data.—Purchasers should specify the following:

(a) Title, number, and date of the specification.

(b) That the responsibility for thorough

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cleaning of containers before filling shall rest with the contractor.

(c) Whether preparation for domestic or overseas shipment is required. (See sec. 5 and 7.2.2.)

6.2 Battery water is not pyrogen-free and must not be used as a vehicle, solvent, or diluent for substances to be administered parenterally.

6.3 Transportation description.

6.3.1 *Transportation description applicable to this item is:*

Water, plain, other than carbonated.
Carload weight 30,000 pounds.
Truckload weight 30,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specification and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standards.*

Navy Department Specification:

51C35—Carboys, Glass, Five-Gallon; and Crates, Tilter-Top.

Military Standard:

MIL–STD–129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Packaging and packing.*

7.2.2.1 *For domestic shipment, immediate use.*—Battery water shall be prepared for shipment in accordance with 5.1 and 5.2.

7.2.2.2 *For domestic shipment and storage or overseas shipment.*—Battery water shall be furnished in 5-gallon capacity glass containers packed for shipment in slatted crates conforming to Navy Department Specification 51C35. Unless otherwise specified in the contract or order, the crates shall be furnished without the tilter top arrangement. Metal or plastic screw cap closures or cork stoppers will be acceptable closing devices.

7.2.3 *Marking.*—In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with Military Standard MIL–STD–129. The nomenclature shall be as follows:

Water, Battery.

MILITARY INTEREST:

Army—C E O Sig T
Navy—Sh A *MC*
Air Force.

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